

Entropy-Minimizing Control of a Helicopter

Santiago Restrepo

contact@waajacu.com <https://waajacu.com/>

Proposed thermodynamic optimal-control method

Abstract. This note proposes entropy-generation-minimizing control for a conventional helicopter near hover. A nonlinear predictive controller seeks to reduce irreversible losses while completing a prescribed flight task. The method combines coupled rotorcraft dynamics, thermodynamic loss maps, and explicit operating constraints; it requires vehicle-specific identification and closed-loop validation.

Index Terms: Helicopter, entropy generation, exergy, nonlinear optimal control.

I. DEFINE THE MISSION AND DYNAMICS

Specify a fixed-duration hover-to-hover maneuver, payload, flight corridor, and endpoint tolerances. This prevents apparent savings obtained by doing less work. Assume a single main rotor, a tail rotor, and governed rotor speed.

$$\begin{aligned} dx/dt &= f(x, u, w), \\ u &= [\theta_0, \theta_{1c}, \theta_{1s}, \theta_1]^T. \end{aligned} \quad (1)$$

The inputs are main collective, two cyclic pitches, and tail collective. State x contains position, velocity, attitude, body rates, and relevant rotor flapping, inflow, actuator, and thermal states; w represents wind. Identify the coupled force and moment model around trim. Include governor dynamics when rotor-speed transients matter.

II. ACCOUNT FOR PHYSICAL ENTROPY

Choose a boundary covering the aircraft, propulsion, and specified wake relaxation. Define entropy generation rate σ in W/K. At fixed ambient temperature T_0 , destroyed exergy equals T_0 times generated entropy [1]. A component model gives

$$\begin{aligned} \sigma &= \sum_j P_{\text{irr},j}/T_j + \sigma_{\text{other}} \geq 0, \\ P_{\text{ex,dest}} &= T_0 \sigma. \end{aligned} \quad (2)$$

Here $P_{\text{irr},j}$ is locally dissipated power at absolute temperature T_j ; σ_{other} includes heat-transfer, mixing, and chemical or electrochemical irreversibility as applicable. Fit these terms from component balances and measured loss maps without double counting.

Rotor-induced power initially leaves as organized wake energy. Count it as entropy generation only when wake relaxation is included; otherwise retain it as outgoing exergy. Shaft power divided by ambient temperature is therefore not a general entropy model. Track stored mechanical and thermal energy so the horizon cannot hide losses.

III. CONNECT ROTOR LOADS TO THE COST

A near-hover power estimate for each rotor is [2]

$$P_r \approx \kappa_r F_r^{3/2} / (2\rho A_r)^{1/2} + P_{\text{profile},r}. \quad (3)$$

F_r is thrust, A_r disk area, ρ air density, and κ_r the induced-power correction. Sum main- and tail-rotor requirements and drivetrain losses. Use calibrated inflow and drag maps for moving flight; the hover relation does not cover stall or vortex-ring descent. These power terms feed the entropy accounting in Section II.

IV. SOLVE A CONSTRAINED FLIGHT PROBLEM

Discretize the dynamics over N steps of duration Δt and minimize total generated entropy:

$$\begin{aligned} \min_U J &= \sum_{k=0}^{N-1} \Delta t \sigma(x_k, u_k, w_k), \\ x_{k+1} &= f_{\Delta t}(x_k, u_k, w_k), \\ e_k \in E_k, x_N \in X_f, (x_k, u_k, \Delta u_k) &\in C. \end{aligned} \quad (4)$$

U is the input sequence, x_0 the estimated state, and w_k the wind forecast. E_k bounds tracking error; X_f specifies terminal flight and mechanical/thermal states. C limits pitch commands and rates, rotor speed, shaft torque/power, temperatures, attitude, clearance, and the identified flight envelope. Choose a terminal set compatible with a stabilizing controller; entropy minimization alone does not ensure stability.

V. IMPLEMENT AND VALIDATE

First identify the model and solve offline by direct multiple shooting or collocation (e.g., CasADi [3]). Then estimate states, warm-start the optimization, apply its first input, and repeat as nonlinear model predictive control. Check feasibility and computation time; retain a stabilizing fallback.

Compare the same maneuver with PID/LQR under wind, mass, and model errors. Report tracking error, integrated entropy or a declared power proxy, input energy, peak torque, and violations. Hover still requires positive power; no efficiency gain is established without these tests.

REFERENCES

- [1] F. Philipp *et al.*, "Optimal control of port-Hamiltonian systems: energy, entropy, and exergy," 2024. arXiv:2306.08914.
- [2] W. Johnson, *NDARC: NASA Design and Analysis of Rotorcraft*, NASA/TP-2015-218751, 2015, ch. 11.
- [3] CasADi, "Optimal control with CasADi," documentation, sec. 8. web.casadi.org/docs.